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surface integral of magnetic field over a closed surface is zero. Mathematically,

$$\oint \vec{B} \cdot d\vec{A} = 0 \qquad \text{(No name)} \qquad (5.8)$$

where $\vec{B}$ is the magnetic field. This equation implies that the magnetic lines of force form a continuous closed path. In other words, it means that no isolated magnetic monopole exists.

3. Third equation is Faraday's law of electromagnetic induction. This law relates electric field with the changing magnetic flux which is mathematically written as

$$\oint \vec{E} \cdot d\vec{l} = -\frac{d}{dt}\Phi_B \text{(Faraday's law)} \qquad (5.9)$$

where $\vec{E}$ is the electric field. This equation implies that the line integral of the electric field around any closed path is equal to the rate of change of magnetic flux through the closed path bounded by the surface. Our modern technological revolution is due to Faraday's laws of electromagnetic induction. The electrical energy supplied to our houses from electricity board by using Faraday's law of induction.

4. Fourth equation is modified Ampere's circuital law. This is also known as Ampere – Maxwell's law. This law relates the magnetic field around any closed path to the conduction current and displacement current through that path.

$$\oint \vec{B}.\overrightarrow{dl} = \mu_0 I_{enclosed} + \mu_0 \varepsilon_0 \frac{d}{dt}\int_s \vec{E}.\overrightarrow{dA} \qquad (5.10)$$

(Ampere-Maxwell's law)

where $\vec{B}$ is the magnetic field. This equation shows that both conduction and also displacement current produces magnetic field. These four equations are known as Maxwell's equations in electrodynamics. This equation ensures the existence of electromagnetic waves. The entire communication system in the world depends on electromagnetic waves. In fact our understanding of stars, galaxy, planets etc come by analysing the electromagnetic waves emitted by these astronomical objects.

5.2 ELECTROMAGNETIC WAVES

Electromagnetic waves are non-mechanical waves which move with speed equals to the speed of light (in vacuum). It is a transverse wave. In the following subsections, we discuss the production of electromagnetic waves and its properties, sources of electromagnetic waves and also classification of electromagnetic spectrum.

5.2.1 Production and properties of electromagnetic waves - Hertz experiment

Maxwell's prediction was experimentally confirmed by Heinrich Rudolf Hertz (Figure 5.7 (a)) in 1888. The experimental set up used is shown in Figure 5.7 (b).

It consists of two metal electrodes which are made of small spherical metals as shown in Figure 5.7. These are connected to larger spheres and the ends of them are connected to induction coil with very large number of turns. This is to produce very high electromotive force (emf). Since the

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